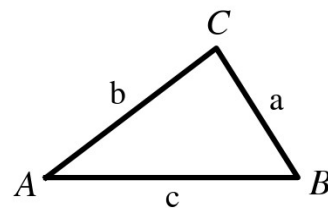


Speak Like A Mathematician:

oblique triangle:

labeling convention:



Law of Sines and Law of Cosines

Law of Sines: $a / \sin A = b / \sin \underline{\hspace{1cm}} = c / \sin \underline{\hspace{1cm}}$. Use it with **AAS, ASA** (two angles + any side).

Law of Cosines: $a^2 = b^2 + c^2 - 2bc \cdot \cos \underline{\hspace{1cm}}$. Use it with **SAS** (two sides + included angle) and **SSS**.

Area formula: $\text{Area} = \frac{1}{2} \cdot a \cdot b \cdot \sin(\underline{\hspace{1cm}})$ — half the product of two sides times the sine of the **INCLUDED** angle.

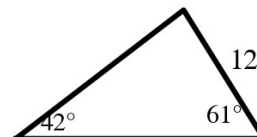
Example 1 - Law of Sines with AAS

In the triangle at the right, $m\angle A = 42^\circ$, $m\angle B = 61^\circ$, and $a = 12$. Round sides to the nearest tenth.

a) Find $m\angle C$.

b) Find b .

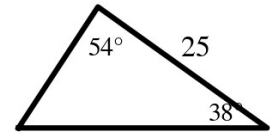
c) Find c .



Example 2 - Law of Sines with ASA

In the triangle at the right, $m\angle B = 38^\circ$, $m\angle C = 54^\circ$, and the side between them (side a) is 25.

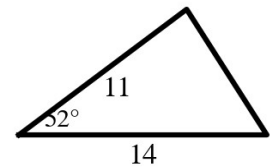
- a) Find $m\angle A$, then b and c (nearest tenth).



Example 3 - Law of Cosines with SAS

In the triangle at the right, $b = 11$, $c = 14$, and the included angle A measures 52° .

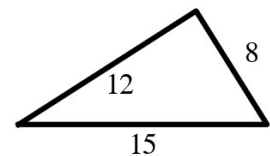
- a) Why can't the Law of Sines start this problem?
- b) Find a (nearest tenth).



Example 4 - Law of Cosines with SSS

The triangle at the right has sides 8, 12, and 15.

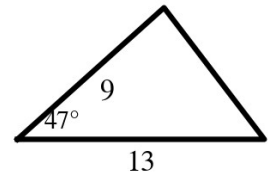
- a) Find the LARGEST angle (nearest tenth). Which side is it opposite?
- b) How did the negative cosine warn you about the answer?



Example 5 - Area Without an Altitude

The triangle at the right has sides 9 and 13 with a 47° included angle.

- a) Find the area (nearest tenth).
- b) Why does the formula need the INCLUDED angle?



Example 6 - Choosing the Law

- a) Given two angles and any side — which law?
- b) Given two sides and the included angle — which law?
- c) Given all three sides — which law, and what comes out first?
- d) Given a right triangle — do you need either law?